

Writing Small Numbers in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Lesson 9 — Writing Small Numbers in Scientific Notation	Date	
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.4

2. Standards

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

Must. Express very small quantities in scientific notation and read them back; determine how many times larger or smaller one number is than another.

Must not. Coefficients are at least 1 and less than 10. A negative exponent never makes the value negative, and operations in scientific notation are Unit 4.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — negative powers of ten are set against the decimal places, so 10^{-3} is read as three places to the right of the point.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Board 7 — a comparison of two very small numbers; students say which exponent wins before they look at the coefficients.
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Board 9 and the bank — thicknesses, tolerances and densities, which are the quantities this notation exists for.

4. Learning Target and Essential Question

Learning target. I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

Essential question. How does a negative power of 10 let us write a very small number, and compare it to another?

Building on. 3.04 — the negative exponent law, $10^{-n} = 1/10^n$; 3.08 — scientific notation for large numbers; Grade 5 — decimal place value to the thousandths.

Working toward. MA.8.NSO.1.5 (Unit 4) — adding, subtracting, multiplying and dividing in scientific notation; MA.8.NSO.1.6 — real-world problems in scientific notation, whose clarification brings in significant digits.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
0.003 becomes 3×10^{-3} , and 3×10^3 is 3,000. Why does the same digit 3 give numbers so far apart?	DOK 2
A student says 0.0000988 is 9.88×10^{-4} because there are four zeros. Where does the count actually come from?	DOK 3
Two numbers have the same coefficient and exponents of -3 and -7 . How many times larger is the first, and how do you know without dividing the coefficients?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	10^{-7}	0.0000001	0.00000001 — seven zeros written, then the 1 (eight places) [8.NSO.1.4: the exponent counts places]; $-10,000,000$ — negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception]
2	Write 0.003 in scientific notation.	3×10^{-3}	3×10^3 — the sign was dropped: that is 3,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]; 3×10^{-2} — counted the zeros instead of the place
3	Write 2.5×10^{-4} in standard form.	0.00025	0.000025 — point moved five places [8.NSO.1.4: 10^{-4} moves the point four places]; 25,000 — point moved right instead of left [negative exponent read as a large number]
4	A machine's cutting tolerance is 0.0000988 inch. The part it cuts is 4 inches long. Write the tolerance in scientific notation.	9.88×10^{-5}	9.88×10^{-4} — counted the zeros [8.NSO.1.4: exponent = places moved]; 988×10^{-7} — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]; anything built from 4 — the unneeded figure was used
5	Which is 0.00081 written in scientific notation?	D — 8.1×10^{-4}	(A) counted the three zeros instead of the four places [8.NSO.1.4: exponent = places moved]; (B) the sign was dropped: 8.1×10^4 is 81,000 [8.NSO.1.4: numbers less than 1 need a negative exponent]; (C) equal in value, but 81 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]
6	How many times larger is 6×10^{-3} than 2×10^{-5} ?	A — 300	(B) 10^2 read as 10 [B1G-M 8.NSO.1.4 misconception: comparing two numbers in scientific notation incorrectly]; (C) coefficients only — the powers were ignored [8.NSO.1.4: both factors are compared]; (D) the exponents were subtracted the wrong way, $-5 - (-3) = -2$ [8.NSO.1.3: quotient law, top minus bottom]
7	Which is greater, 5.3×10^{-4} or 8.9×10^{-6} ?	5.3×10^{-4}	8.9×10^{-6} — coefficients compared first, or -6 read as larger than -4 [8.NSO.1.4: compare the powers of 10 first, and $-6 < -4$]
8	A computer displays a number as $4.2E-6$. Write the number in standard form.	0.0000042	$-4,200,000$ — the exponent's sign read as the number's sign [B1G-M 8.NSO.1.4 misconception: confusing the exponent with the value of the number]; 0.00000042 — point moved seven places
9	Fishing line A is 2×10^{-2} inch thick. Fishing line B is 6×10^{-2} inch thick. Which is thicker, and how many times thicker? Give a possible thickness between the two.	B; 3 times; e.g. 4×10^{-2} in.	' 2×10^{-2} is thicker' — 10^{-2} read as making a smaller coefficient bigger; ' 4×10^{-1} in between' — power ignored in the in-between value [8.NSO.1.4: both factors matter]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the decimal-place strip from Notes I — 10^{-1} through 10^{-8} with the places marked — and let the student walk the decimal point along it, one place per power, rather than counting zeros in their head.

ELL. The comparison questions turn on two phrases, times larger and times smaller. Put both on the board over the same pair of numbers and read the question out loud twice before any board goes up.

Enrichment. Ask how many times larger 5×10^{-2} is than 2×10^{-6} , then for a number between 4×10^{-5} and 5×10^{-5} written in scientific notation.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name B as the thicker line, give the factor 3, and write a thickness that falls between the two. Read the boards, not the papers.

Vocabulary. **negative power of 10; scientific notation**

Materials. Whiteboards and markers. Calculators are allowed on everything; entering 4.2×10^{-6} and reading the display back as 4.2E-6 is worth thirty seconds at whiteboard 8.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Convert between standard and scientific notation: small numbers — 85M; Compare small numbers — KDR; Powers of ten with negative exponents — SKD.